

Lecture 18: CYK Algorithm

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CYK Algorithm: The algorithm to decide whether a string belongs to the language of a grammar or not.

Basics of CYK Algorithm:

- ✓ The Structure of the rules in a Chomsky Normal Form grammar
- ✓ Uses a “dynamic programming” or “table-filling approach”

Construct a triangular table:

Each row corresponds to one length of substrings

- ✓ Bottom Row – Strings of length 1
- ✓ Second from Bottom Row – Strings of length 2
- ✓ Top Row – string ‘w’
- $X_{i,i}$ is the set of variables A such that
 $A \rightarrow w_i$ is a production of G
- Compare at most n pairs of previously computed sets:
 $(X_{i,i}, X_{i+1,j}), (X_{i,i+1}, X_{i+2,j}) \dots (X_{i,j-1}, X_{j,j})$

$X_{1,5}$				
$X_{1,4}$	$X_{2,5}$			
$X_{1,3}$	$X_{2,4}$	$X_{3,5}$		
$X_{1,2}$	$X_{2,3}$	$X_{3,4}$	$X_{4,5}$	
$X_{1,1}$	$X_{2,2}$	$X_{3,3}$	$X_{4,4}$	$X_{5,5}$
w_1	w_2	w_3	w_4	w_5

Table for string ‘w’ that has length 5

Example: Show the CYK Algorithm with the following example:

– CNF grammar **G**

$S \rightarrow AB \mid BC$

$A \rightarrow BA \mid a$

$B \rightarrow CC \mid b$

$C \rightarrow AB \mid a$

– **w** is baaba

Question Is **baaba** in $L(G)$?

{B}	{A, C}	{A, C}	{B}	{A, C}
b	a	a	b	a

Calculating the Bottom ROW

- $X_{1,2} = (X_{i,i}, X_{i+1,j}) = (X_{1,1}, X_{2,2})$
- $\rightarrow \{B\}\{A,C\} = \{BA, BC\}$
- Steps:
 - Look for production rules to generate BA or BC
 - There are two: S and A
 - $X_{1,2} = \{S, A\}$

{S, A}				
{B}	{A, C}	{A, C}	{B}	{A, C}
b	a	a	b	a

$S \rightarrow AB \mid BC$
 $A \rightarrow BA \mid a$
 $B \rightarrow CC \mid b$
 $C \rightarrow AB \mid a$

- $X_{2,3} = (X_{i,i}, X_{i+1,j}) = (X_{2,2}, X_{3,3})$
- $\rightarrow \{A, C\}\{A, C\} = \{AA, AC, CA, CC\} = Y$
- Steps:
 - Look for production rules to generate Y
 - There is one: B
 - $X_{2,3} = \{B\}$

{S, A}	{B}			
{B}	{A, C}	{A, C}	{B}	{A, C}
b	a	a	b	a

- $X_{3,4} = (X_{i,i}, X_{i+1,j}) = (X_{3,3}, X_{4,4})$
- $\rightarrow \{A, C\}\{B\} = \{AB, CB\} = Y$
- Steps:
 - Look for production rules to generate Y
 - There are two: S and C
 - $X_{3,4} = \{S, C\}$

{S, A}	{B}	{S, C}		
{B}	{A, C}	{A, C}	{B}	{A, C}
b	a	a	b	a

- $X_{4,5} = (X_{i,i}, X_{i+1,j}) = (X_{4,4}, X_{5,5})$
- $\rightarrow \{B\}\{A, C\} = \{BA, BC\} = Y$
- Steps:
 - Look for production rules to generate Y
 - There are two: S and A
 - $X_{4,5} = \{S, A\}$

{S, A}	{B}	{S, C}	{S, A}	
{B}	{A, C}	{A, C}	{B}	{A, C}
b	a	a	b	a

- $X_{1,3} = (X_{i,i}, X_{i+1,j}) (X_{i,i+1}, X_{i+2,j})$
 $= (X_{1,1}, X_{2,3}), (X_{1,2}, X_{3,3})$
- $\rightarrow \{B\}\{B\} \cup \{S, A\}\{A, C\} = \{BB, SA, SC, AA, AC\} = Y$
- Steps:
 - Look for production rules to generate Y
 - There are NONE: S and A
 - $X_{1,3} = \emptyset$
 - no elements in this set (empty set)

$\emptyset$				
{S, A}	{B}	{S, C}	{S, A}	
{B}	{A, C}	{A, C}	{B}	{A, C}
b	a	a	b	a

$S \rightarrow AB \mid BC$

$A \rightarrow BA \mid a$

$B \rightarrow CC \mid b$

$C \rightarrow AB \mid a$

- $X_{2,4} = (X_{i,i}, X_{i+1,j}) (X_{i,i+1}, X_{i+2,j})$
 $= (X_{2,2}, X_{3,4}), (X_{2,3}, X_{4,4})$
- $\rightarrow \{A, C\}\{S, C\} \cup \{B\}\{B\} = \{AS, AC, CS, CC,$
- Steps:
 - Look for production rules to generate Y
 - There is one: B
 - $X_{2,4} = \{B\}$

$\emptyset$	$\{B\}$			
$\{S, A\}$	$\{B\}$	$\{S, C\}$	$\{S, A\}$	
$\{B\}$	$\{A, C\}$	$\{A, C\}$	$\{B\}$	$\{A, C\}$
b	a	a	b	a

- $X_{3,5} = (X_{i,i}, X_{i+1,j}) (X_{i,i+1}, X_{i+2,j})$
 $= (X_{3,3}, X_{4,5}), (X_{3,4}, X_{5,5})$
- $\rightarrow \{A, C\}\{S, A\} \cup \{S, C\}\{A, C\}$
 $= \{AS, AA, CS, CA, SA, SC, CA, CC\} = Y$
- Steps:
 - Look for production rules to generate Y
 - There is one: B
 - $X_{3,5} = \{B\}$

$\emptyset$	$\{B\}$	$\{B\}$		
$\{S, A\}$	$\{B\}$	$\{S, C\}$	$\{S, A\}$	
$\{B\}$	$\{A, C\}$	$\{A, C\}$	$\{B\}$	$\{A, C\}$
b	a	a	b	a

$\{S, A, C\}$	$\leftarrow X_{1,5}$			
$\emptyset$	$\{S, A, C\}$			
$\emptyset$	$\{B\}$	$\{B\}$		
$\{S, A\}$	$\{B\}$	$\{S, C\}$	$\{S, A\}$	
$\{B\}$	$\{A, C\}$	$\{A, C\}$	$\{B\}$	$\{A, C\}$
b	a	a	b	a

- Table for string 'w' that has length 5
- The algorithm populates the triangular table

- Is baaba in $L(G)$?

Yes

We can see the S in the set X_{1n} where 'n' = 5

We can see the table

the cell $X_{15} = (S, A, C)$ then

if $S \in X_{15}$ then baaba $\in L(G)$

- The CYK Algorithm correctly computes X_{ij} for all i and j; thus w is in $L(G)$ if and only if S is in X_{1n} .
- The running time of the algorithm is $O(n^3)$.